Lab 4 - Advanced Machine Learning

Nonnegative Matrix Factorization

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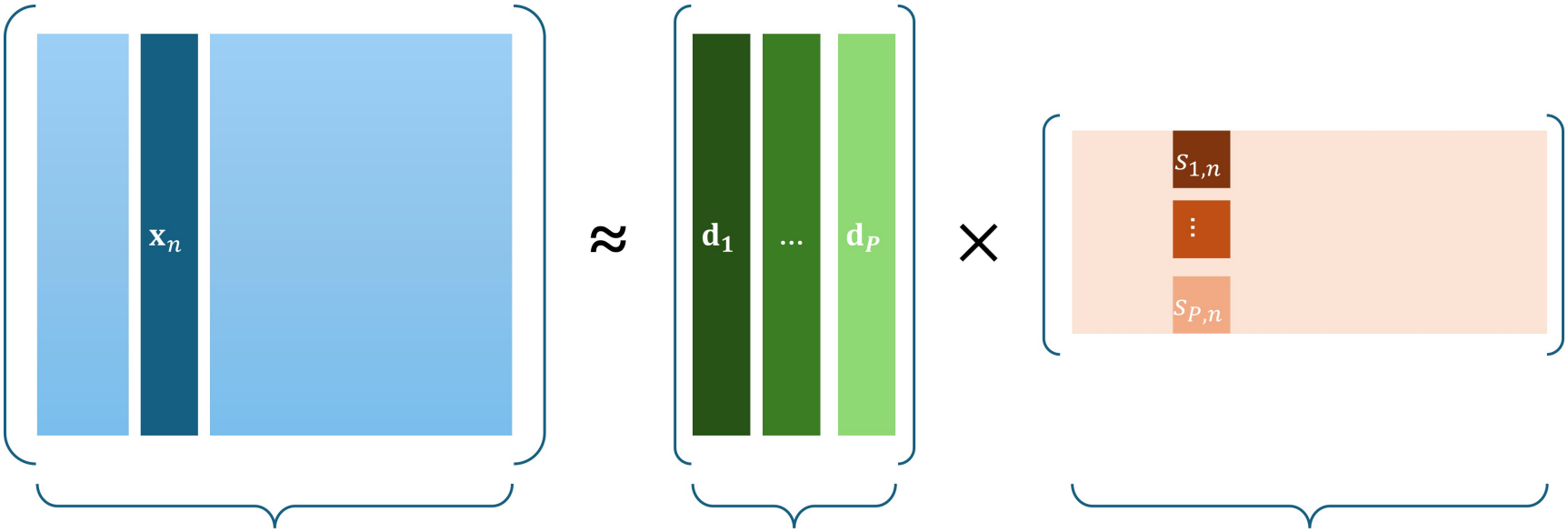
CentraleSupélec

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General Informations

- **Assignment:** You will work **in pairs**.
- **Due date: Each assignment must be sent within 7 days of the lab session (by the next Thursday evening).**
  Send your notebook to `ludovic.trautmann@inria.fr` (only one submission per pair, remember to put both of your names on it).
- **Grading:** There are 6 lab sessions. The average grade will count for one third of your final grade.
  One part of the final exam will cover the material from the labs.

Lesson Recap



N = Number of samples
M = Size of the observed vector
P = Number of features in the dictionary

Each sample $\mathbf{x}_n$ decomposes as a non-negative weighted sum of the features in the dictionary :

$$(\forall n \in [\![1, N]\!]) \quad \mathbf{x}_n \approx \sum_{p=1}^{P} s_{p,n} \mathbf{d}_p = \mathbf{D}\mathbf{s}_n \tag{1}$$

Algorithms : (Regularized) Frobenius Loss

Loss

$$\mathcal{L}_{Frob}(\mathbf{W}, \mathbf{H}) = \frac{1}{2}\|\mathbf{X} - \mathbf{W}\mathbf{H}\|_F^2 + \frac{\mu}{2}\|\mathbf{H}\|_F^2 + \lambda\|\mathbf{H}\|_1 + \frac{\nu}{2}\|\mathbf{W}\|_F^2$$

$$\mathbf{X} \in \mathbb{R}_+^{p \times n}, \quad \mathbf{W} \in \mathbb{R}_+^{p \times r}, \quad \mathbf{H} \in \mathbb{R}_+^{r \times n}$$

Algorithm

$$\mathbf{W} \leftarrow \mathbf{W} \odot \frac{\mathbf{X}\mathbf{H}^\top}{\mathbf{W}(\mathbf{H}\mathbf{H}^\top + \nu \mathbf{I}_r)}$$

$$\mathbf{H} \leftarrow \mathbf{H} \odot \frac{\mathbf{W}^\top \mathbf{X}}{(\mathbf{W}^\top \mathbf{W} + \mu \mathbf{I}_r)\mathbf{H} + \lambda \mathbb{1}_{r \times n}}$$

Algorithms : KL-Loss

Loss

$$\mathcal{L}_{KL}(\mathbf{W}, \mathbf{H}) = \mathcal{KL}(\mathbf{X}, \mathbf{WH})$$

Algorithm

$$W_{ik} \leftarrow W_{ik} \frac{\sum_{l=1}^{n} \frac{H_{kl} X_{il}}{[\mathbf{WH}]_{il}}}{\sum_{l=1}^{n} H_{kl}}$$

$$H_{kj} \leftarrow H_{kj} \frac{\sum_{i=1}^{p} \frac{W_{ik} X_{ij}}{[\mathbf{WH}]_{ij}}}{\sum_{i=1}^{p} W_{ik}}$$